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Institute for Natural Resources Technology and Management

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# Potential improvements in agricultural systems: Insights from WaPOR

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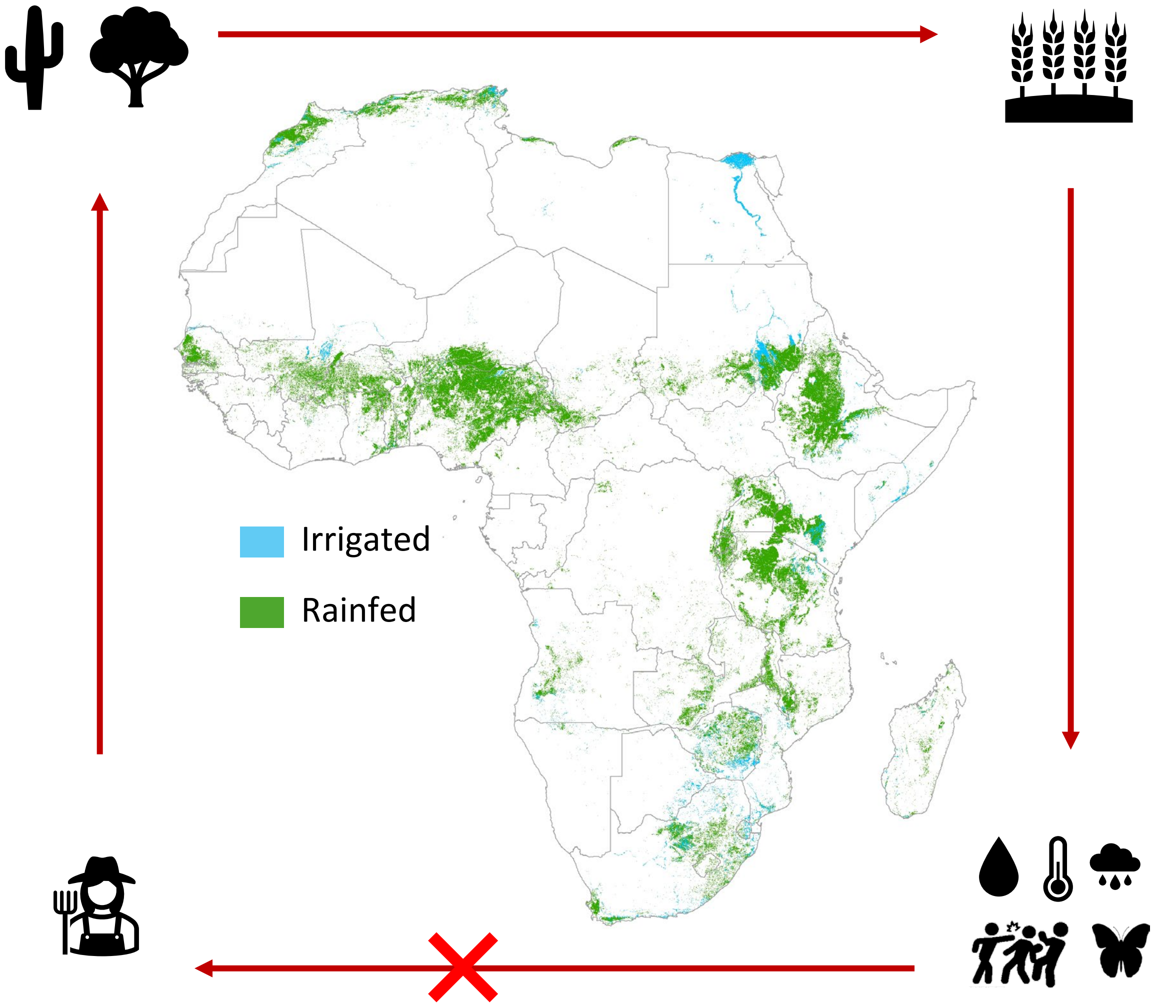
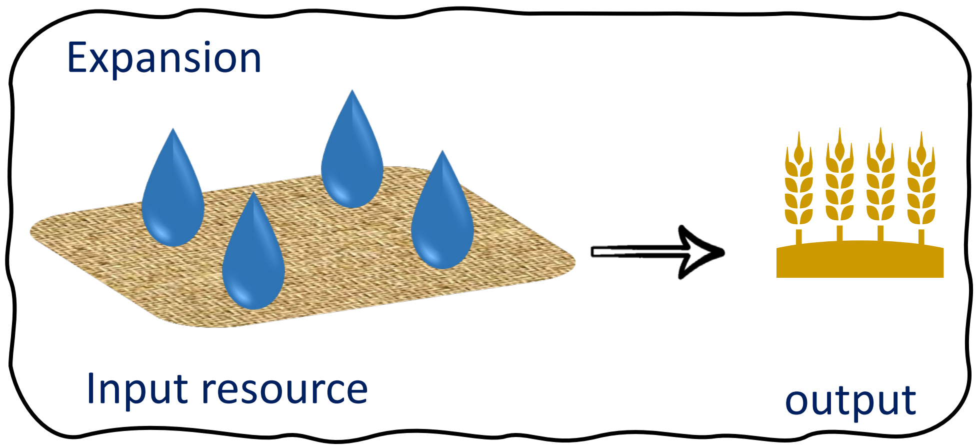
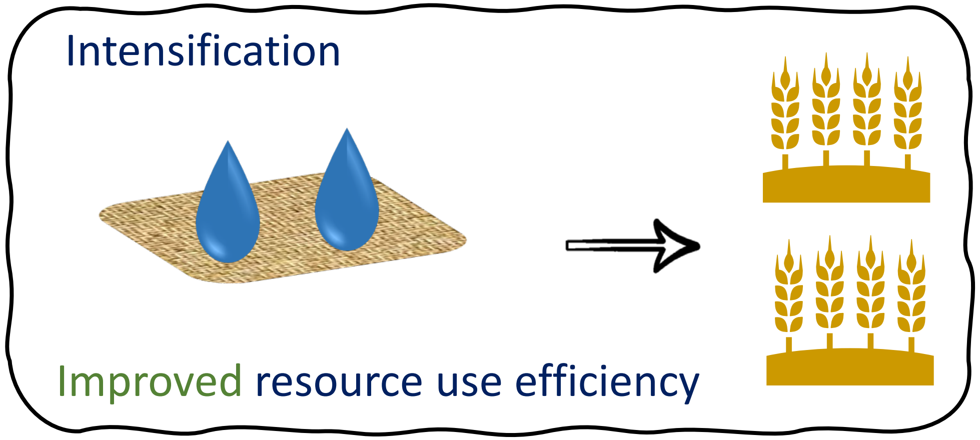
**Improving before expanding!**






- Which crop/season?
- Where?
- Increase output (production)
- Conserve input (water & land)
- Production efficiency indicators
- Resources availability & demand (also in the future!)

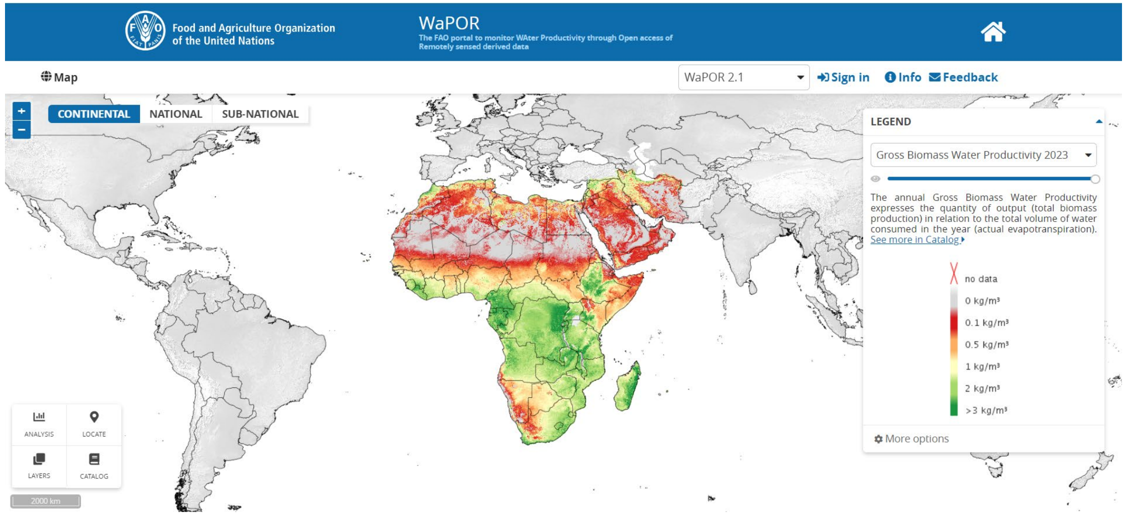
## Good datasets:

- Open-access → knowledge for all, verification & reproduction
- Resolution & coverage → realize objectives
- Continuity → replication & operationalization
- Performance quality in the study region → Important to check

# WaPOR - FAO's portal to monitor Water Productivity through Open access of Remotely sensed derived data

## Version 2

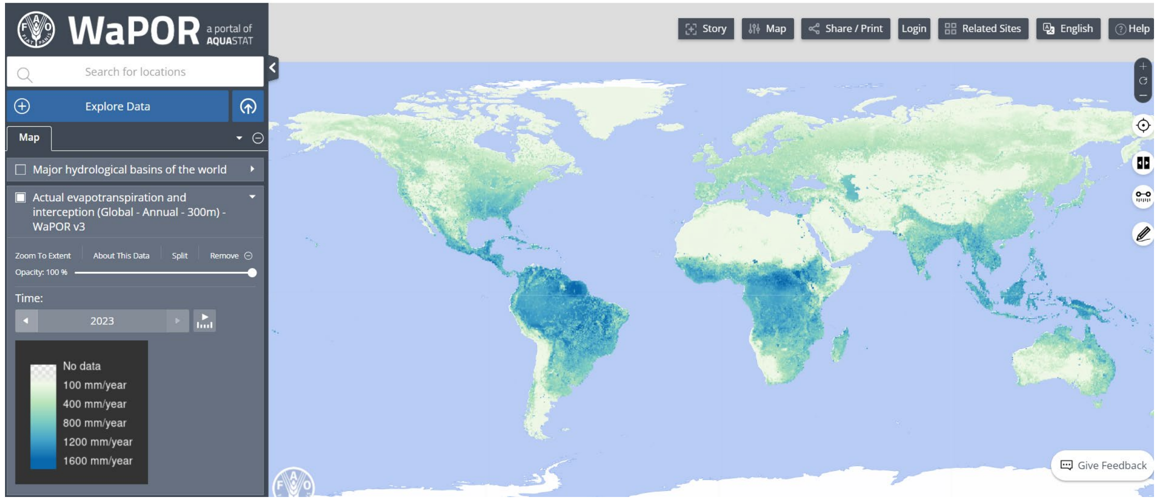
- → 2009 - 2023
- → Africa & Near East (250m)
- → Selected sites (30m)




Egypt - Ethiopia - Mali - Mozambique - Rwanda - Kenya - Sri Lanka - Senegal - Sudan - Libya - Tunisia - Lebanon - Iraq - Yemen

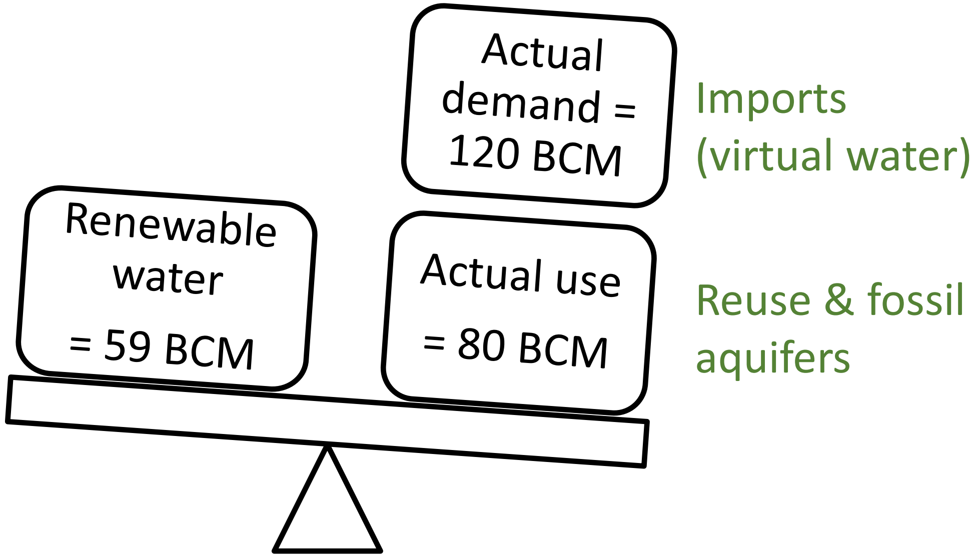
## Version 3

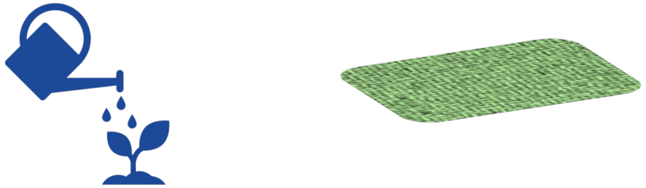
- → 2018 - present
- → Global coverage (300m)
- → Selected sites (20m)




Additionally in V3: Algeria - Colombia - Jordan - Morocco - Pakistan - Palestine - Vietnam

# Potential improvements in crop production in Egypt



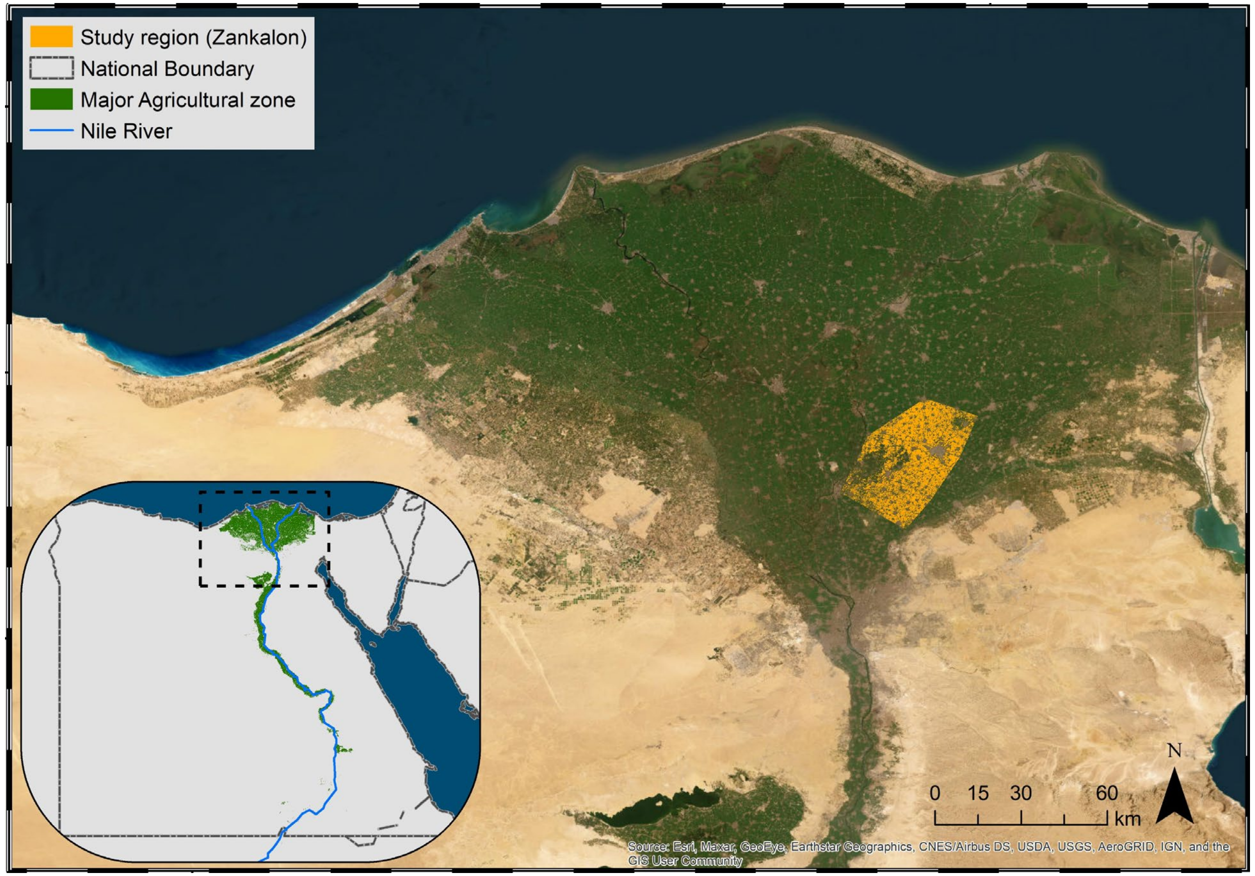



| | | |
|---|---|---|
| **2025** | 62 BCM | 7.0 Mha |
| **2050** | 110 BCM | 9.2 Mha |

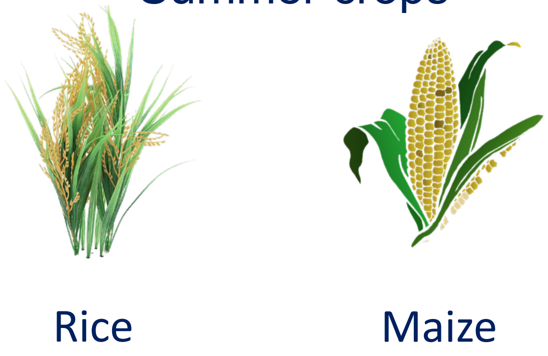
**Ayyad, S.**, Karimi, P., Ribbe, L. and Becker, M., 2024. Potential Improvements in Crop Production in Egypt and Implications for Future Water and Land Demand. International Journal of Plant Production. https://doi.org/10.1007/s42106-024-00301-7

# Zankalon - Nile Delta

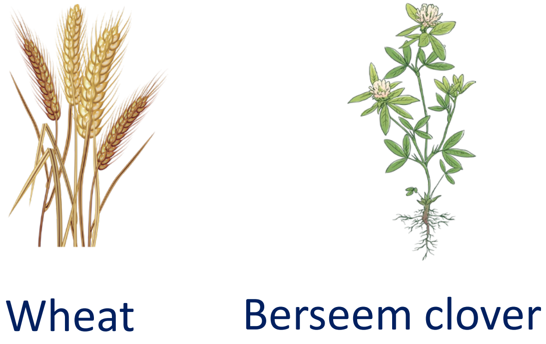
- Hot spot
- Data availability (WaPOR, 30m; 10-day)
- Representativeness




Summer crops




Winter crops




- Main food & feed crops
- 50% of annually harvested areas and renewable freshwater!

# Efficiency Indicators

$$\text{Transpiration fraction} = \frac{\text{Transpiration}}{\text{Actual ET}}$$

$$\text{Crop Water Productivity} = \frac{\text{Crop production}}{\text{Actual ET}}$$

$$\text{Crop Yield} = \frac{\text{Crop production}}{\text{Land area}}$$

- Absolue values
- Potential improvements:

Average values across all fields compared to top 5% performing fields during 2016 - 2021

# Absolute values

Summer

Winter




| | Rice | Maize | Wheat | Berseem |
|---|---|---|---|---|
| Transpiration fraction (%) | 79 | 79 | 83 | 85 |
| Crop water productivity (kg/$m^3$) | 0.2 | 0.3 | 1.05 | 1.2 |
| Crop yield (ton/ha) | 1.5 | 1.8 | 6.3 | 9.2 |

→ Reasonable transpiration fraction values for all crops

→ Reasonable CWP and yield values for winter crops; substantial underestimation for summer crops

→ Relying on the relative than the absolute values

# Potential improvements

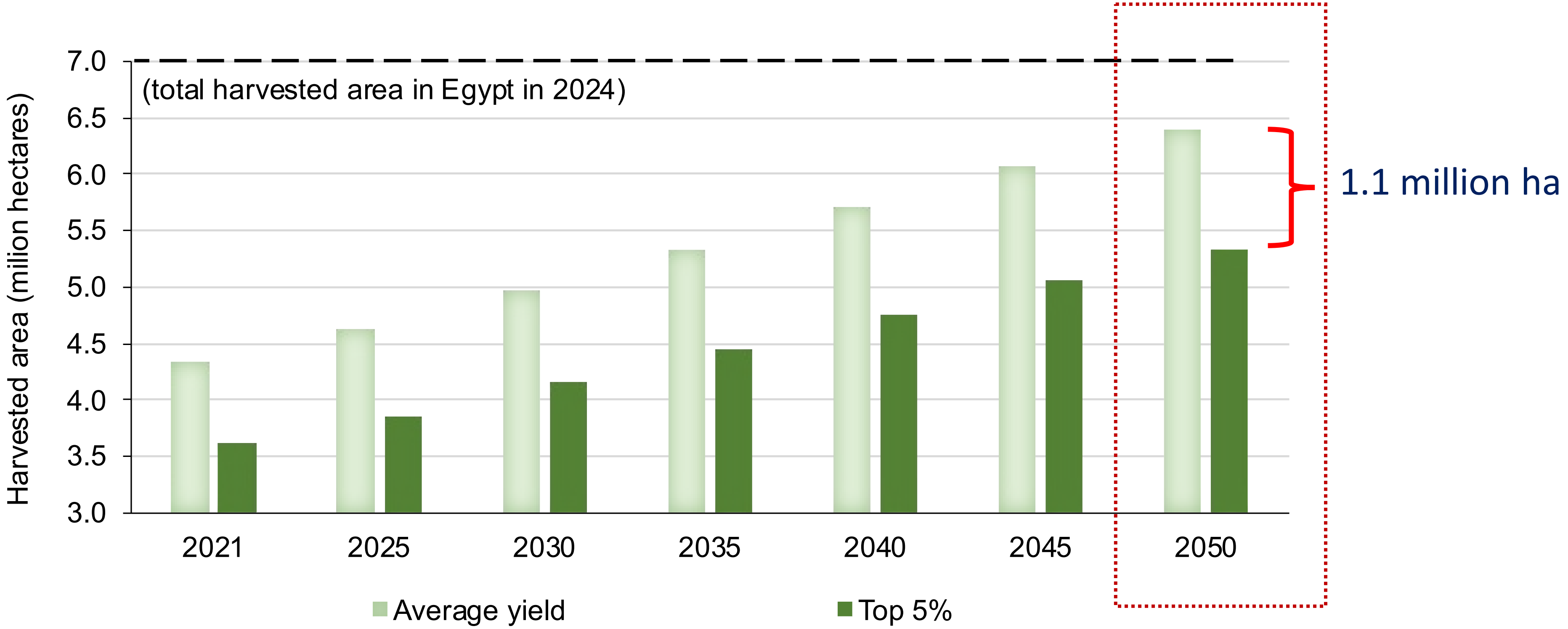
| | Summer | | Winter | |
|---|---|---|---|---|
| | Rice | Maize | Wheat | Berseem |
| Transpiration fraction (%) | 4 | 4 | 2 | 2 |
| Crop water productivity (%) | 14 | 13 | 10 | 9 |
| Crop yield (%) | 27 | 22 | 17 | 16 |

→ Limited potential for transpiration fraction, much higher for crop yield and water productivity

→ Higher gains for summer crops than for winter crops

→ Future land & water demand based on possible improvements and historical domestic crop production per capita

# Projected harvest area



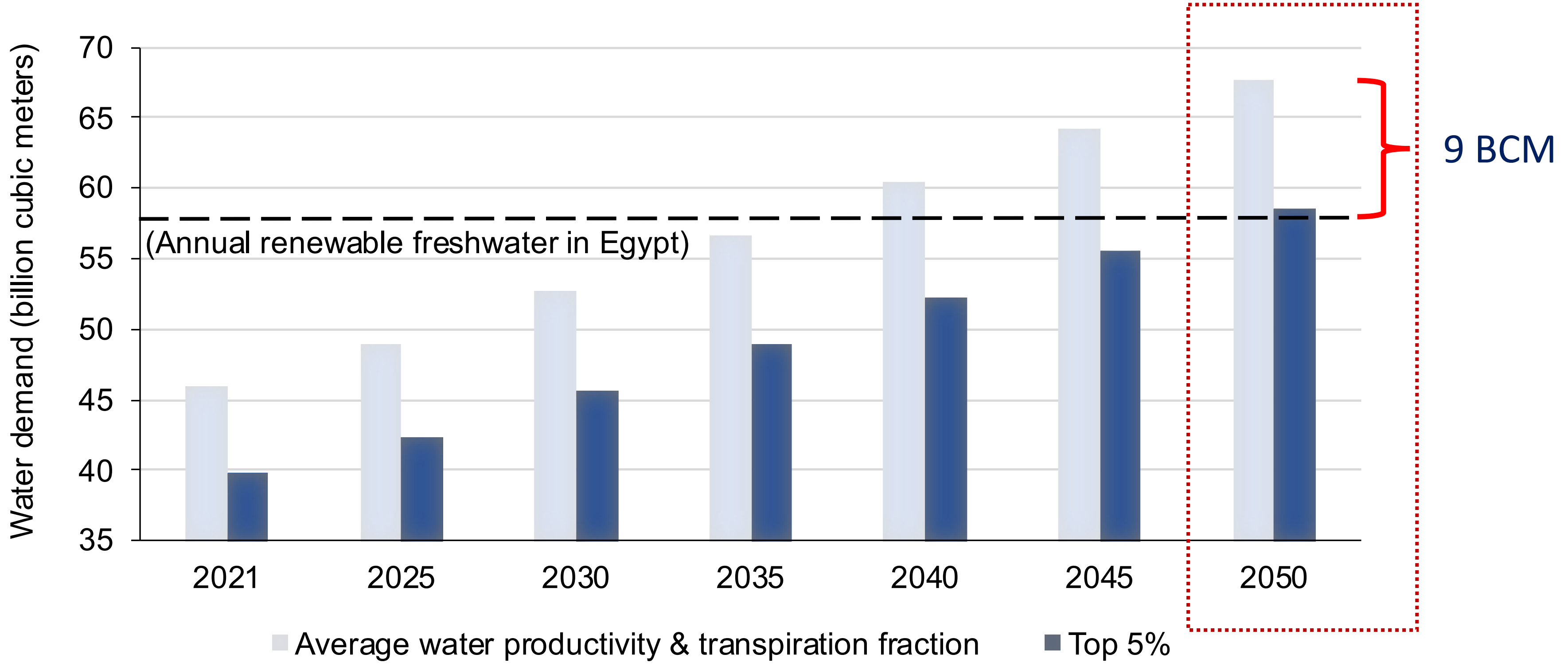

18%

28%

36%

18%

# Projected water demand




# Efforts & investments towards:

## Improving efficiency

→ Water use and canopy coverage are already quite efficient

→ Water productivity can improve, primarily via yield gains

→ Better crop and soil management practices (rotation, cultivars, raised beds, fertilizers, etc)

→ Field & conveyance efficiencies

## Broader interventions

→ Meat imports instead of fodder cultivation

→ Curbing demand (population growth, changes in dietary habits, crop substitution)

- ✓ Which crop/season?
- ✓ Where?
- ✓ Increase output (production)?
- ✓ Conserve input (water & land)?
- ✓ Future demand vs availability?

## Moving forward

→ Regular crop mapping from pilot areas across the country

→ Upscaling indicator assessment to the national scale (more crops; additional sites)

→ Operationalization (web-based dashboard for regular monitoring)

→ Monitor fields before and after interventions: ensure efficiency gains lead to real water savings, not overuse

→ Go beyond physical metrics: why some fields perform better? who may benefit from saved water?

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International Journal of Plant Production (2024) 18:313–334
https://doi.org/10.1007/s42106-024-00301-7


RESEARCH

**Potential Improvements in Crop Production in Egypt and Implications for Future Water and Land Demand**


Saher Ayyad[1,2] · Poolad Karimi[3] · Lars Ribbe[2] · Mathias Becker[1]


Takeaways:

- Improving before expanding
- Novel approaches & WaPOR data can identify potential improvements and entry points
- There is potential for improved efficiency in Egypt


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Agricultural Water Management
Volume 269, 1 July 2022, 107712


**Remote sensing assessment of available green water to increase crop production in seasonal floodplain wetlands of sub-Saharan Africa**


Saher Ayyad [a b], Poolad Karimi [c], Matthias Langensiepen [a], Lars Ribbe [b], Lisa-Maria Rebelo [d], Mathias Becker [a]



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Journal of Hydrology: Regional Studies
Volume 37, October 2021, 100884


**How well do gridded precipitation and actual evapotranspiration products represent the key water balance components in the Nile Basin?**


Ian McNamara [a], Oscar M. Baez-Villanueva [a b], Ali Zomorodian [a], Saher Ayyad [a c], Mauricio Zambrano-Bigiarini [d e], Modathir Zaroug [f], Azeb Mersha [g], Alexandra Nauditt [a], Milly Mbuliro [f], Sowed Wamala [f], Lars Ribbe [a]



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Science of The Total Environment
Volume 775, 25 June 2021, 145769


**Will the Eastern Nile countries be able to sustain their crop production by 2050? An outlook from water and land perspectives**


Saher Ayyad [a c], Muhammad Khalifa [a b]



earth
MDPI


*Article*

**Satellite-Based Approach for Crop Type Mapping and Assessment of Irrigation Performance in the Nile Delta**


Samar Saleh [1,*], Saher Ayyad [1,2] and Lars Ribbe [1]